SET-3

Series AABB2/4

प्रश्न-पत्र कोड
Q.P. Code **56/4/3**

रोल नं.
Roll No.

परीक्षार्थी प्रश्न-पत्र कोड को उत्तर-पुस्तिका के मुख-पृष्ठ पर अवश्य लिखें ।
Candidates must write the Q.P. Code on the title page of the answer-book.

- कृपया जाँच कर लें कि इस प्रश्न-पत्र में मुद्रित पृष्ठ **11** हैं ।
- प्रश्न-पत्र में दाहिने हाथ की ओर दिए गए प्रश्न-पत्र कोड को परीक्षार्थी उत्तर-पुस्तिका के मुख-पृष्ठ पर लिखें ।
- कृपया जाँच कर लें कि इस प्रश्न-पत्र में **12** प्रश्न हैं ।
- **कृपया प्रश्न का उत्तर लिखना शुरू करने से पहले, उत्तर-पुस्तिका में प्रश्न का क्रमांक अवश्य लिखें ।**
- इस प्रश्न-पत्र को पढ़ने के लिए 15 मिनट का समय दिया गया है । प्रश्न-पत्र का वितरण पूर्वाह्न में 10.15 बजे किया जाएगा । 10.15 बजे से 10.30 बजे तक छात्र केवल प्रश्न-पत्र को पढ़ेंगे और इस अवधि के दौरान वे उत्तर-पुस्तिका पर कोई उत्तर नहीं लिखेंगे ।
- Please check that this question paper contains **11** printed pages.
- Q.P. Code given on the right hand side of the question paper should be written on the title page of the answer-book by the candidate.
- Please check that this question paper contains **12** questions.
- **Please write down the serial number of the question in the answer-book before attempting it.**
- 15 minute time has been allotted to read this question paper. The question paper will be distributed at 10.15 a.m. From 10.15 a.m. to 10.30 a.m., the students will read the question paper only and will not write any answer on the answer-book during this period.

रसायन विज्ञान (सैद्धान्तिक)

CHEMISTRY (Theory)

निर्धारित समय : 2 घण्टे | *अधिकतम अंक : 35*

Time allowed : 2 hours | *Maximum Marks : 35*

General Instructions :

Read the following instructions very carefully and strictly follow them :

(i) *This question paper contains **12** questions. **All** questions are compulsory.*

(ii) *This question paper is divided into **three** Sections – **A, B** and **C**.*

(iii) ***Section A** – Questions no. **1** to **3** are very short answer type questions, carrying **2** marks each.*

(iv) ***Section B** – Questions no. **4** to **11** are short answer type questions, carrying **3** marks each.*

(v) ***Section C** – Question no. **12** is case based question, carrying **5** marks.*

(vi) *Use of log tables and calculators is **not** allowed.*

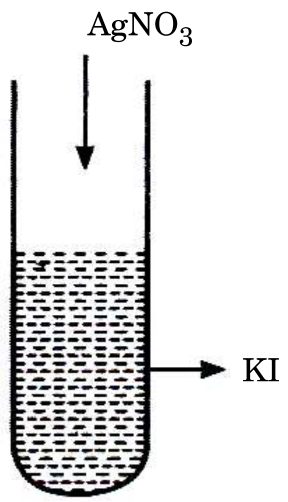
SECTION A

1. Write reasons for the following statements : *2×1=2*

(i) Benzoic acid does not undergo Friedel-Crafts reaction.

(ii) Oxidation of aldehydes is easier than that of ketones.

2. Observe the given figure and answer the following questions : (Any ***two***) *2×1=2*



(i) What is the charge on AgI colloidal particles ?

(ii) What is the reason for the origin of the charge on AgI ?

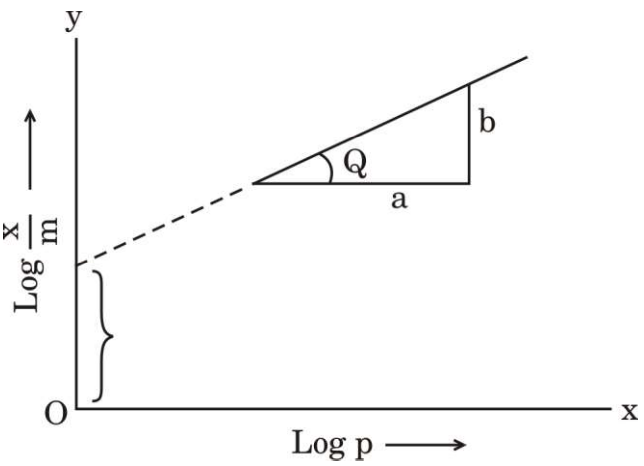
(iii) If KI is added to $AgNO_3$, what will be the charge on AgI colloidal particles ?

3. (i) $(CH_3)_3C - CHO$ does not undergo aldol condensation. Why ?

(ii) Distinguish between Acetophenone and Benzophenone with the help of a chemical test. *1+1=2*

SECTION B

4. Observe the given figure and answer the following questions : *3×1=3*



(i) Write the expression for adsorption of gases on solids in the form of an equation.

(ii) What is the slope of the graph ?

(iii) What does the intercept of the line represent ?

5. (a) (i) Write the electronic configuration of d^4 on the basis of crystal field splitting theory if $\Delta_o > P$.

(ii) $[Co(NH_3)_6]^{3+}$ is an inner orbital complex whereas $[Ni(NH_3)_6]^{2+}$ is an outer orbital complex.
(Atomic number : Co = 27, Ni = 28)

(iii) Write the number of ions produced from the complex $[Pt(NH_3)_6]Cl_4$ in solution. *3×1=3*

OR

(b) (i) Calculate the spin only magnetic moment of the complex $[Fe(H_2O)_6]^{2+}$. (Atomic number of Fe = 26)

(ii) Which out of the following two complexes is more stable and why ?

$[CoF_6]^{3-}$, $[Co(C_2O_4)_3]^{3-}$

(iii) Write the IUPAC name of the given complex : *3×1=3*

$[Pt(NH_3)_2Cl_2]$

6. (a) Write equations involved in the following reactions : *3×1=3*

(i) Ethanamine reacts with acetyl chloride.

(ii) Aniline reacts with bromine water at room temperature.

(iii) Aniline reacts with chloroform and ethanolic potassium hydroxide.

OR

(b) (i) Write the IUPAC name for the following organic compound :

$(CH_3CH_2)_2NCH_3$

(ii) Write the equations for the following :

(I) Gabriel phthalimide synthesis

(II) Hoffmann bromamide degradation *1+2=3*

7. (a) Write reasons for the following : *3×1=3*

(i) Ethylamine is soluble in water whereas aniline is insoluble.

(ii) Amino group is *o*- and *p*-directing in aromatic electrophilic substitution reactions, but aniline on nitration gives a substantial amount of *m*-nitroaniline.

(iii) Amines behave as nucleophiles.

OR

(b) How will you carry out the following conversions : *3×1=3*

(i) Nitrobenzene to Aniline

(ii) Ethanamide to Methanamine

(iii) Ethanenitrile to Ethanamine

8. Arrange the following compounds in the increasing order of their property indicated : *3×1=3*

(i) Acetaldehyde, Benzaldehyde, Acetophenone, Acetone (Reactivity towards HCN)

(ii) $(CH_3)_2CHCOOH$, $CH_3CH_2CH(Br)COOH$, $CH_3CH(Br)CH_2COOH$ (Acidic strength)

(iii) CH_3CH_2OH, CH_3CHO, CH_3COOH (Boiling point)

9. (i) Which ion amongst the following is colourless and why ?

Ti^{4+}, Cr^{3+}, V^{3+}

(Atomic number of Ti = 22, Cr = 24, V = 23)

(ii) Why is Mn^{2+} much more resistant than Fe^{2+} towards oxidation ?

(iii) Highest oxidation state of a metal is shown in its oxide or fluoride only. Justify the statement. *3×1=3*

10. Write the Nernst equation and calculate the emf of the following cell at 298 K : *3*

$$Zn \mid Zn^{2+}(0{\cdot}1\ M) \parallel Cd^{2+}(0{\cdot}01) \mid Cd$$

Given : $E^{\ominus}_{Zn^{2+}/Zn} = -\,0{\cdot}76\ V$

$E^{\ominus}_{Cd^{2+}/Cd} = -\,0{\cdot}40\ V$

(log 10 = 1)

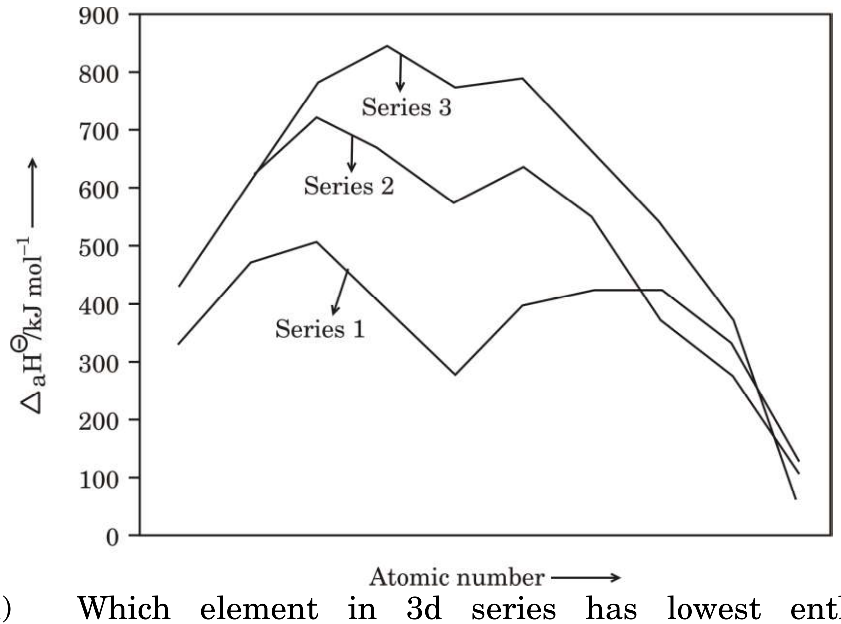
11. (a) (i) Silver atom has completely filled d-orbitals in its ground state, it is still considered to be a transition element. Justify the statement.

(ii) Why are $E^{\ominus}_{M^{2+}/M}$ values of Mn and Zn more negative than expected ?

(iii) Why do transition metals form alloys ? *3×1=3*

OR

(b) Answer the following questions on the basis of the figure given below : *3×1=3*



(i) Which element in 3d series has lowest enthalpy of atomisation ?

(ii) Why do metals of the second and third series have greater enthalpies of atomisation ?

(iii) Why are enthalpies of atomisation of transition metals quite high ?

SECTION C

12. Read the passage given below and answer the questions that follow : *1+1+1+2=5*

The rate law for a chemical reaction relates the reaction rate with the concentrations or partial pressures of the reactants. For a general reaction aA + bB $\rightarrow$ C with no intermediate steps in its reaction mechanism, meaning that it is an elementary reaction, the rate law is given by $r = k[A]^x[B]^y$, where [A] and [B] express the concentrations of A and B in moles per litre. Exponents x and y vary for each reaction and are determined experimentally. The value of k varies with conditions that affect reaction rate, such as temperature, pressure, surface area, etc. The sum of these exponents is known as overall reaction order. A zero order reaction has a constant rate that is independent of the concentration of the reactants. A first order reaction depends on the concentration of only one reactant. A reaction is said to be second order when the overall order is two. Once we have determined the order of the reaction, we can go back and plug in one set of our initial values and solve for k.

(i) Calculate the overall order of a reaction which has the following rate expression : *1*

$$\text{Rate} = k[A]^{1/2}\,[B]^{3/2}$$

(ii) What is the effect of temperature on rate of reaction ? *1*

(iii) What is meant by rate of a reaction ? *1*

(iv) (a) A first order reaction takes 77·78 minutes for 50% completion. Calculate the time required for 30% completion of this reaction. (log 10 = 1, log 7 = 0·8450) *2*

OR

(b) A first order reaction has a rate constant 1×10^{-3} per sec. How long will 5 g of this reactant take to reduce to 3 g ?

(log 3 = 0·4771; log 5 = 0·6990) *2*